Electronics Lab Report

Course: ENGR-UH 3611 Electronics (Fall 2025)

Lab Report: Instrumentation Amplifier (Pre-Lab #4)

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Objective

The objective of this lab is to design, simulate, and analyze an instrumentation amplifier (IA) with a total voltage gain of 500 V/V. The amplifier is implemented in Cadence, using op-amp models with specified parameters, differential input sources, and a capacitive load. The performance of the amplifier is evaluated by comparing the theoretical and simulated gain, and by analyzing transient simulation results.

Theory/Background

Instrumentation amplifiers are precision differential amplifiers widely used in sensor signal conditioning due to their ability to amplify small differential voltages while rejecting large common-mode signals. The general form of an IA consists of three operational amplifiers and a resistor network.

The differential gain of a classic three-op-amp IA is given by:

$$A_v = \left(1 + \frac{2R1}{R_{gain}}\right) * \left(\frac{R3}{R2}\right)$$

where:

- Rgain- controls the overall gain,
- R1, R2, R3 define the resistor ratios in the input and differential amplifier stages.

For this design, the target gain is 500 V/V. With supply voltages of +5 V and -5 V, and a differential input of two sine waves (100 Hz, 0.1 mV amplitude, 180° out of phase). The op-amp models are non-ideal, with parameters:

- Gain = 10^6
- Gain-bandwidth product = 10^8
- Input bias current = 10^{-12} A
- Input impedance = 10^{12} Ω

Design

The gain requirement was achieved by selecting resistor values according to the gain equation.

Resistor	Value (kΩ)
R1	500
R2	20
R3	20
Rgain	2

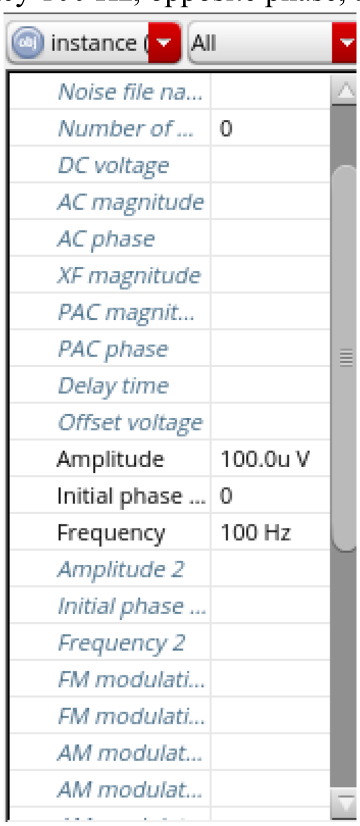
Table 1. Resistor values for instrumentation amplifier design

Gain, Av = (1 + (2(500)/2)) * (20/20) = 501 V/V

Simulation Setup Procedure

The IA was implemented and simulated in Cadence following these steps:

1. Voltage sources: Two sinusoidal sources from analogLib, with amplitude 0.1 mV, frequency 100 Hz, opposite phase, and supply rails at ±5 V.



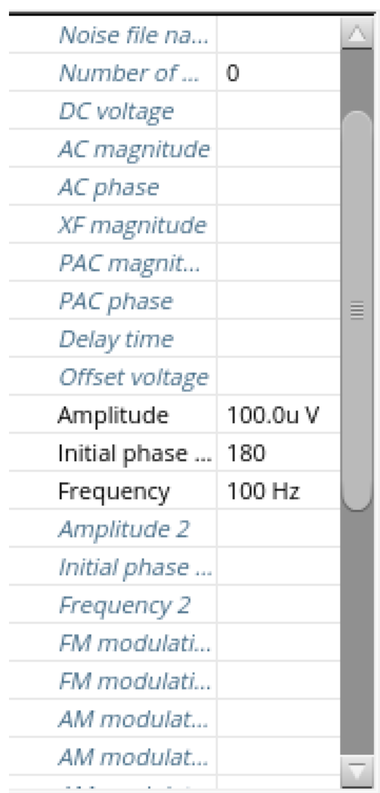


Figure 1.0 Configuration of Input Sinusoidal Signals

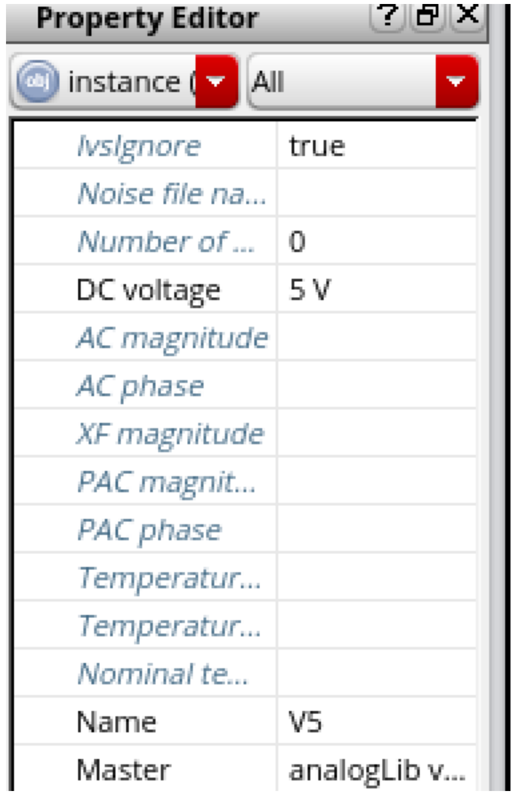


Figure 1.1 Configuration of Power Supply Rails

2. Op-amp models: Configured with the given parameters.

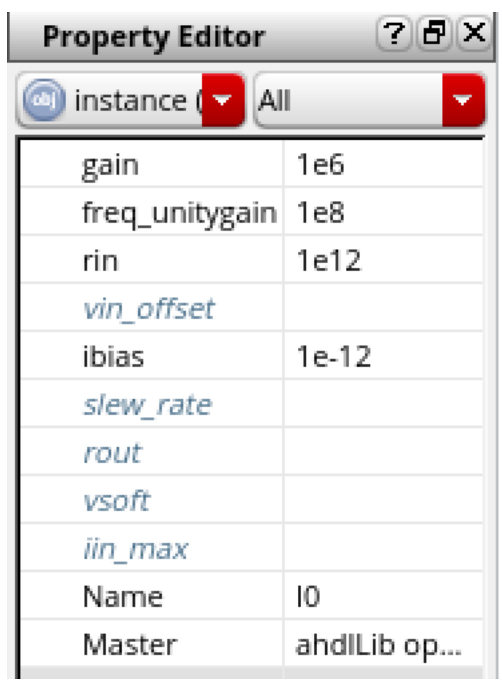


Figure 1.2 Configuration of Operational Amplifiers

3. Load capacitor: 10 pF connected at the output.

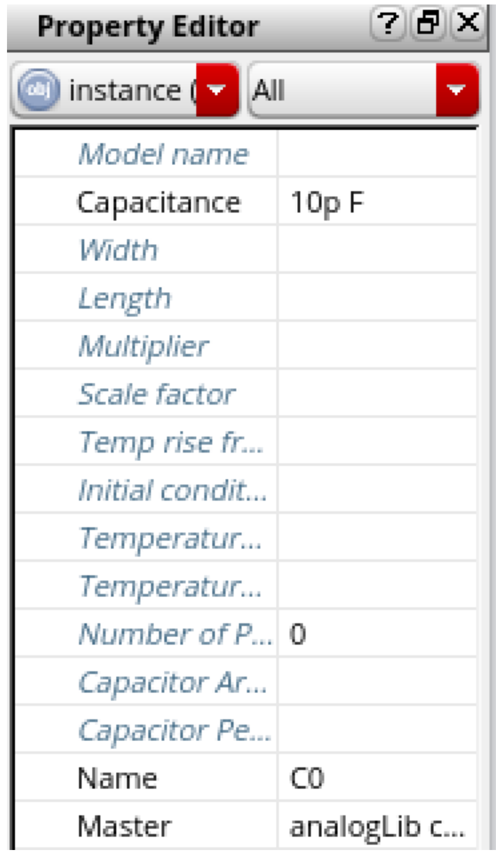


Figure 1.3 Configuration of Capacitor at Output

4. Testbench: Constructed with IA symbol instantiation, input sources, output probes, and supply connections.
5. Simulation: Transient analysis performed to observe input and output waveforms over multiple cycles.

Results

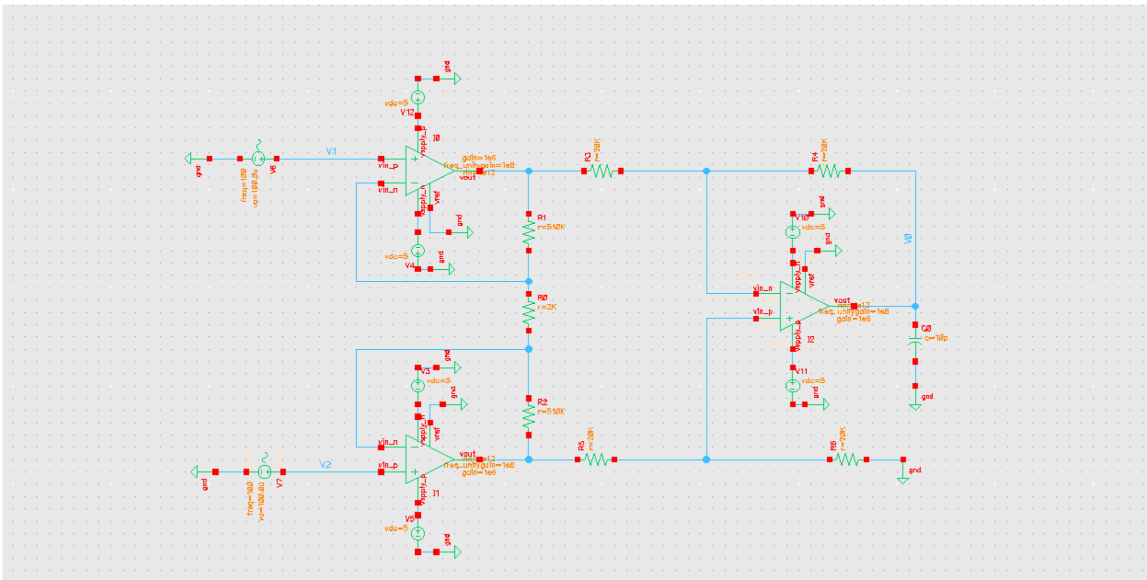


Figure 2. Internal Schematic of Instrumentation Amplifier Design

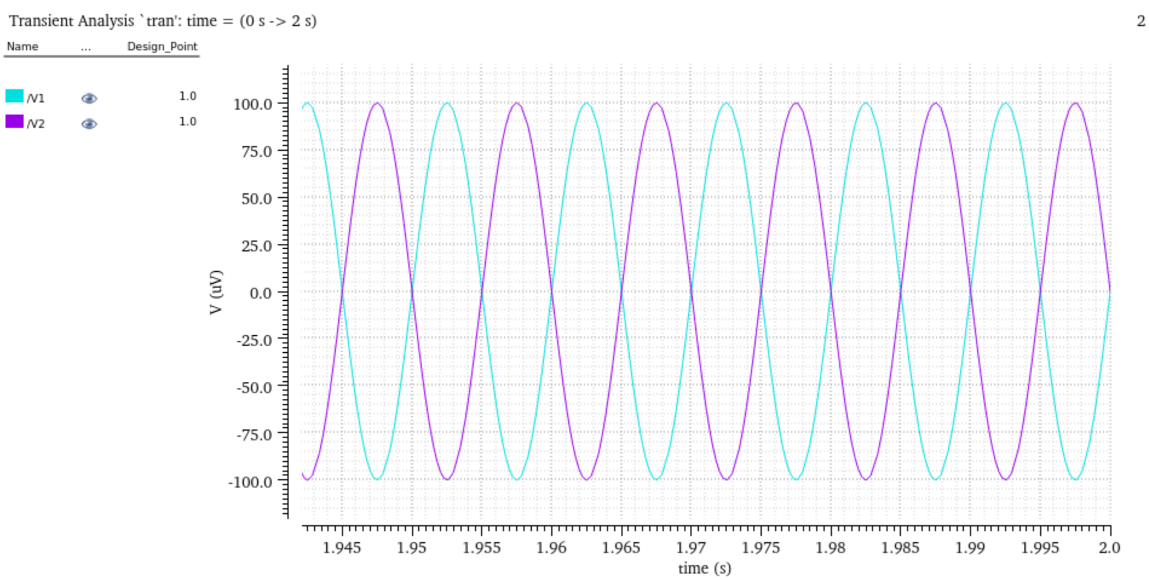


Figure 3. Input Signals Waveform for Simulation of Instrumentation Amplifier

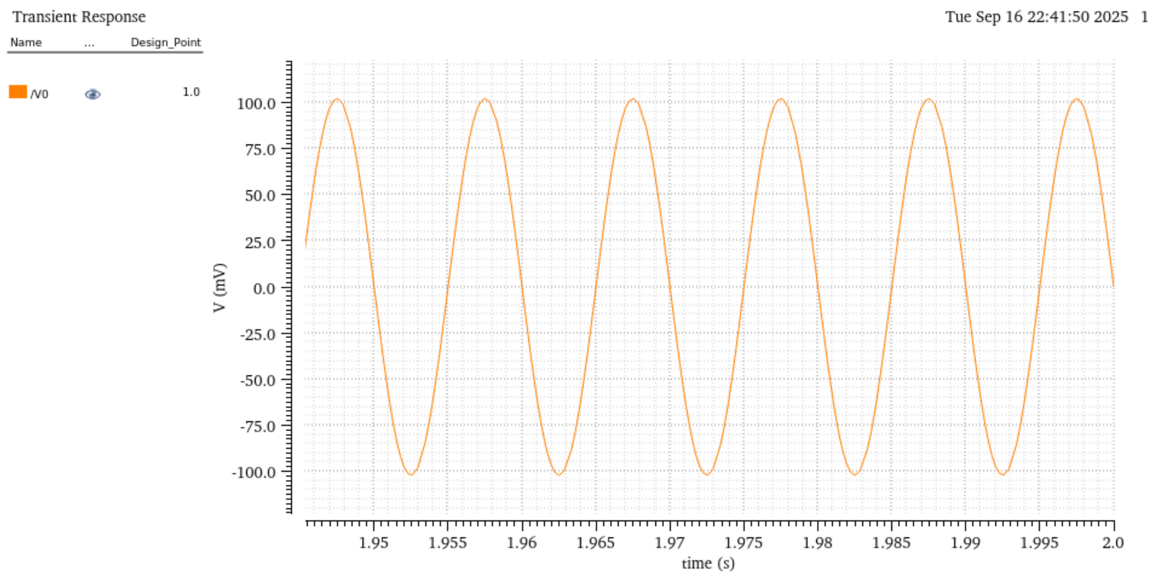


Figure 4. Output Signal Waveform for Simulation of Instrumentation Amplifier

Gain, Av = V0/ (V1-V2) = 100 mV/ (0.1 –(-0.1)) mV = 500 V/V

The simulated waveforms confirm proper operation of the IA. The two input signals were 0.1 mV amplitude producing a differential voltage of 0.2 mV, while the output shows an amplified differential signal of amplitude 100 mV (200 mV peak-to-peak), 100 mV, demonstrating the differential input was amplified by a gain of 500 V/V.

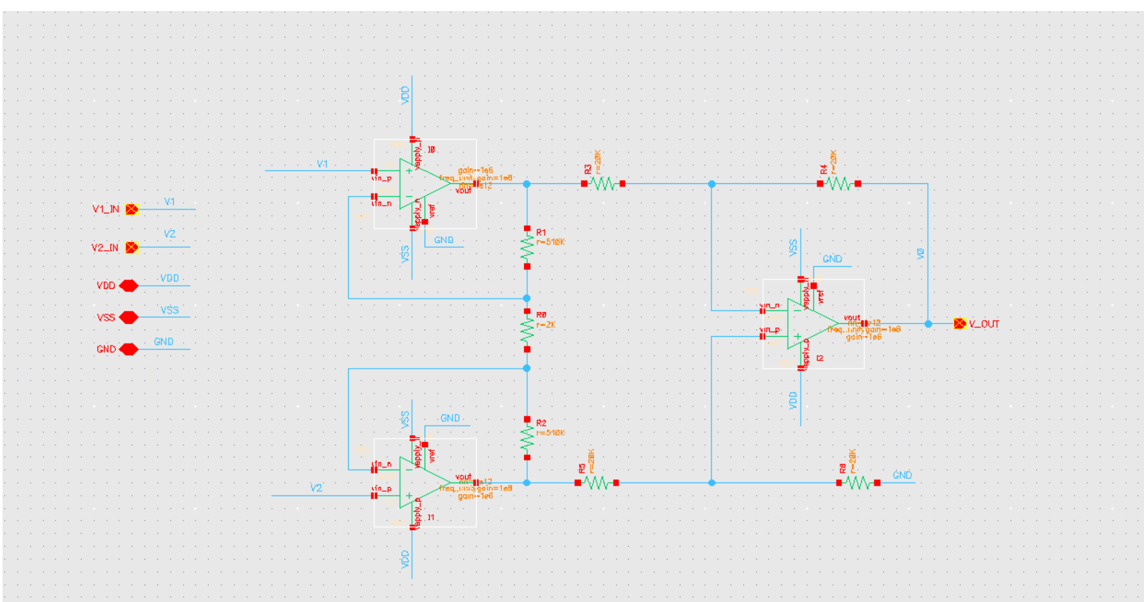


Figure 5. Testbench Schematic with input, output, power(VDD, VSS) and ground pins.

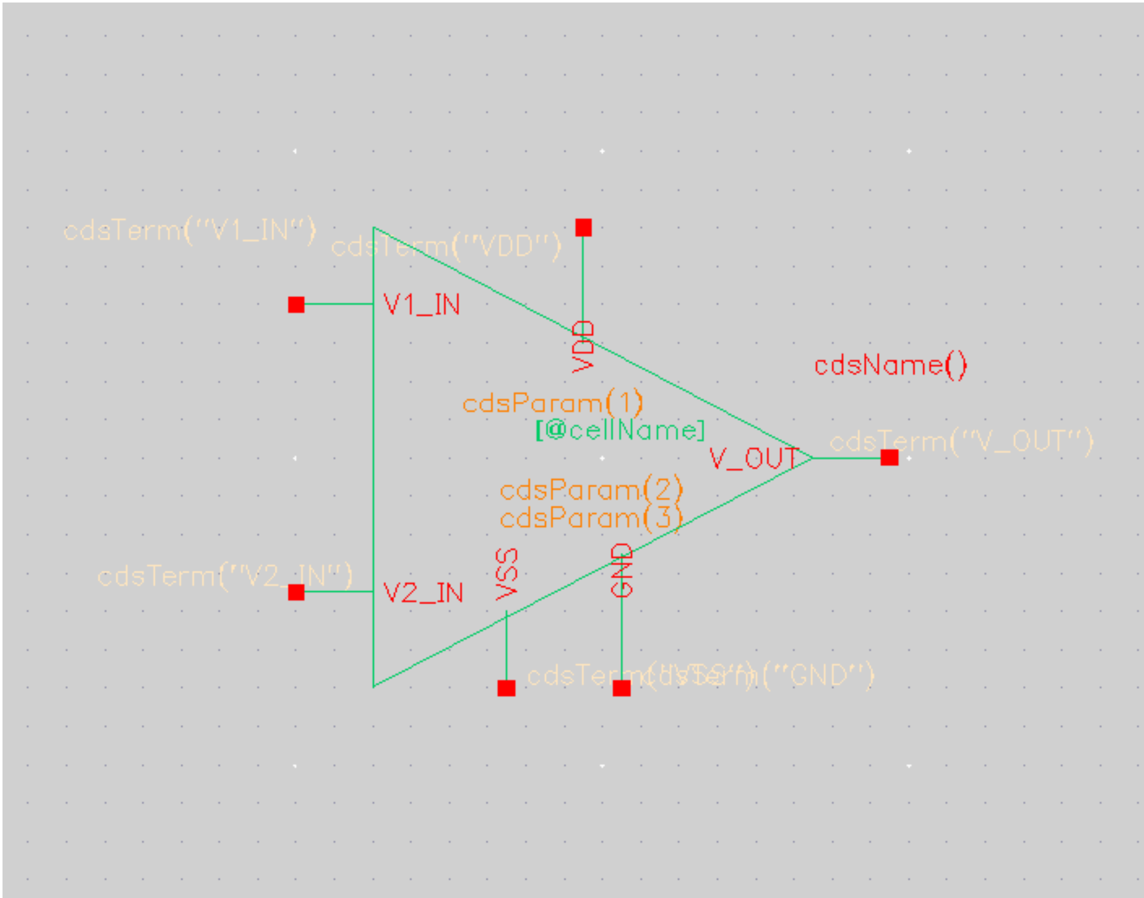


Figure 6. Testbench Symbol Schematic with input, output, power(VDD, VSS) and ground pins.

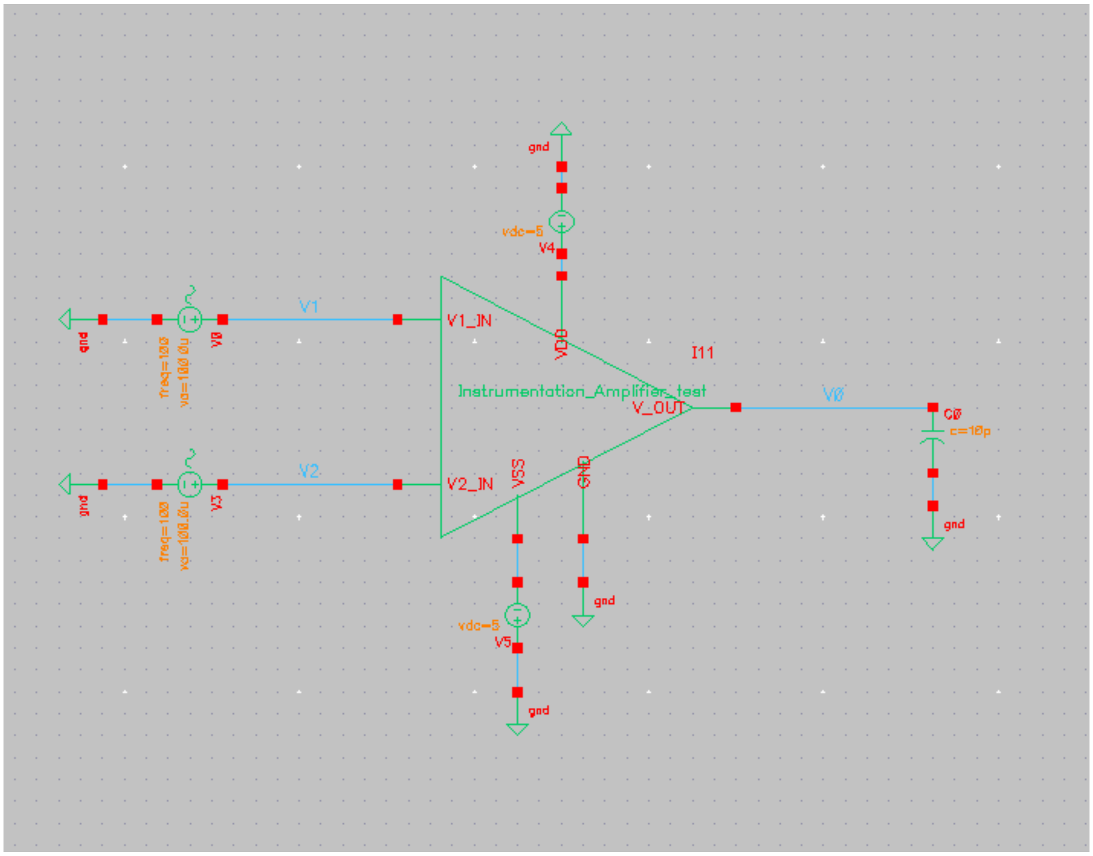


Figure 7. Schematic of Instantiated Instrumentation Amplifier Connected to 10 pF Capacitor

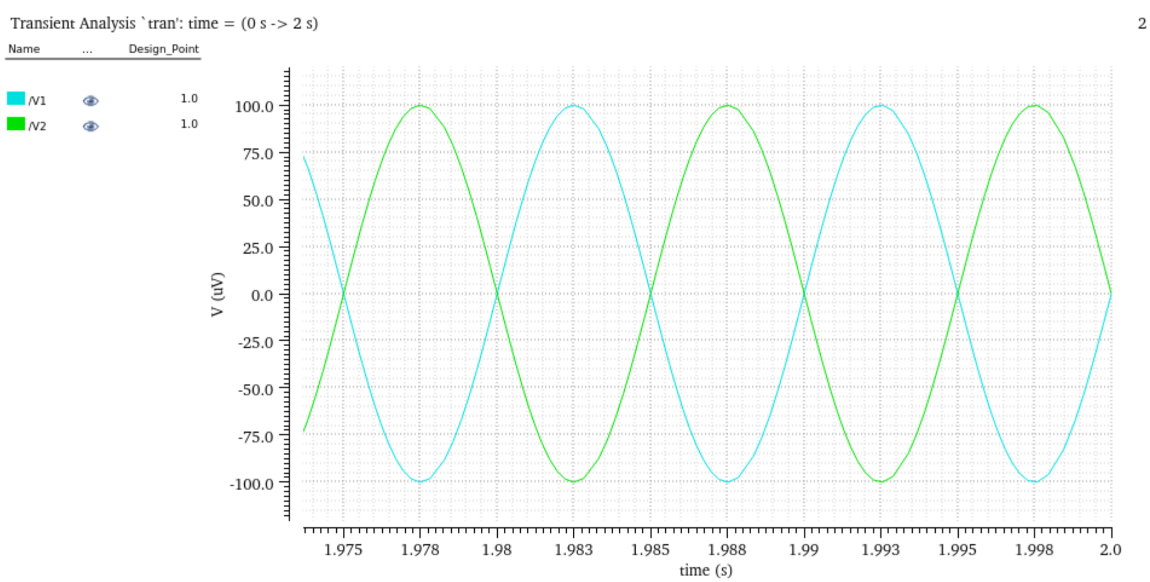


Figure 8. Input Signals Waveform for Simulation of the Instantiated Instrumentation Amplifier

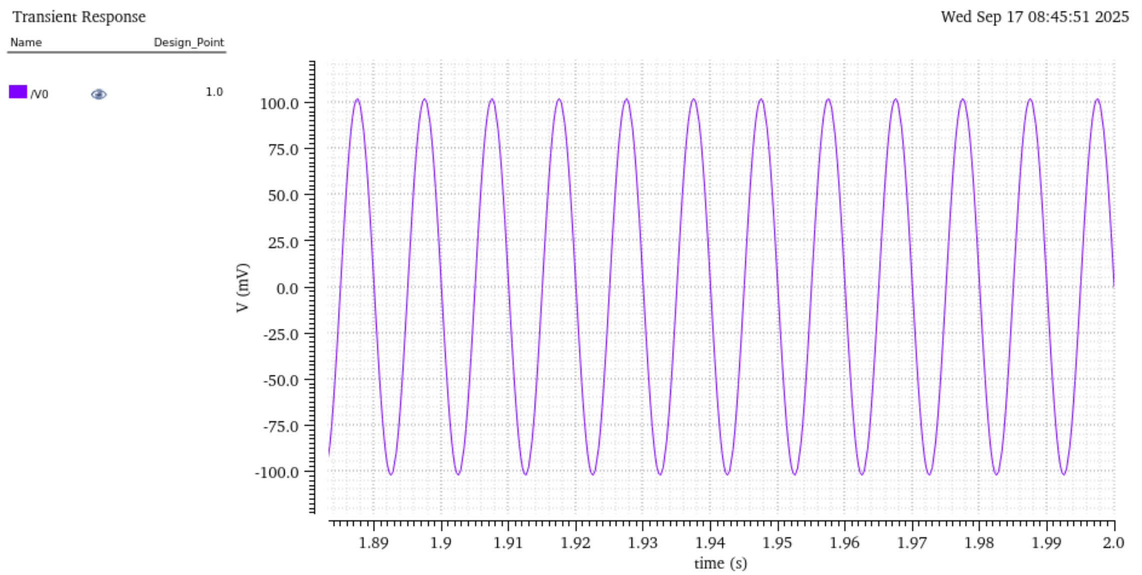


Figure 9. Output Signal Waveform for Simulation of the Instantiated Instrumentation Amplifier

Gain, Av = V0/ (V1-V2) = 100 mV/ (0.1 –(-0.1)) mV = 500 V/V

The simulated waveforms for the instantiated IA confirm proper operation of the IA. The two input signals were 0.1 mV amplitude producing a differential voltage of 0.2 mV, while the output shows an amplified differential signal of amplitude 100 mV (200 mV peak-to-peak), demonstrating the differential input was amplified by a gain of 500 V/V.

Conclusion

The IA was designed for a gain of 500 V/V and implemented successfully in Cadence.

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References

1. Sedra, A. S., & Smith, K. C. Microelectronic Circuits, 7th Edition.
2. Course lecture notes, ENGR-UH 3611.
3. Cadence Analog Design Environment Documentation.